Rohit Sagar Chavan

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PROFESSIONAL SUMMARY

Data Scientist & Analyst with 3+ years of experience in automation, MIS, and predictive analytics. Built ETL pipelines handling 500K+ records daily and automation workflows that save 20+ hours every week. Delivered Power BI dashboards tracking 25+ KPIs for 5+ departments, and built machine learning models (Naive Bayes, Random Forest, XGBoost) that cut unnecessary field visits by 40%. Skilled in Python, SQL, Tableau, Power BI, and advanced Excel. Recently completed a Master's in Computer Science (Machine Learning / AI Engineering)).

PROFESSIONAL EXPERIENCE

Executive Data Analyst Sep 2024 – Present
Meteorology Data Pvt. Ltd. (BARC INDIA Subsidiary) *Mumbai, India*

- Engineered 15+ automation workflows using Python, reducing manual reporting time by 75% and saving 20+ hours weekly.
- Built end-to-end data pipelines for ETL processes handling 500K+ records daily, improving data accuracy by 95%
- Worked on an ML model (Naive Bayes) project to predict viewership and reduce unnecessary field visits by 40%.
- Developed interactive Tableau dashboards tracking 25+ KPIs, enabling data-driven decisions for 5+ departments.
- Led cross-functional collaboration with 8+ teams to align automation strategies, improving process efficiency by 40%.

Executive – MIS and Analytics Jun 2023 – Sep 2024
Adecco *Mumbai, India*

- Designed and implemented SOPs for MIS workflows across 6 departments, standardizing data handling and reporting protocols.
- Created automated email reporting systems delivering 50+ daily reports to stakeholders, ensuring 99% on-time delivery.
- Established a centralized data repository consolidating 10+ data sources, reducing data retrieval time by 60%.
- Analyzed gaps in existing MIS practices and optimized reporting accuracy, reducing data discrepancies by 85%.

Assistant Trainer (Data Science) – Part-time Feb 2024 – Apr 2024
DeveLearn *Mumbai, India*

- Trained students in data analysis using Python, SQL, Advanced Excel, Power BI, and Tableau through hands-on, project-based sessions.
- Taught core Python and SQL concepts, including data cleaning, querying, and building dashboards for reporting.
- Supported learners with doubt-solving and practical assignments to help them apply data analysis skills to real datasets.

KEY PROJECTS

- **IndustryGPT: LLM-Based IT Support Bot (Capstone)**
  Fine-tuned Flan-T5-small (Hugging Face) on a cleaned corpus of 17,158 IT helpdesk Q&A pairs, on CPU only. Raised ROUGE-1 from 0.131 to 0.331 over the untrained baseline. Built a retrieval layer (TF-IDF, cosine similarity) that returns real historical answers first and falls back to the model, with an ethical-request filter. Deployed as a Streamlit chat app.
- **DeepFER: Facial Emotion Recognition (Deep Learning, Capstone)**
  Built a 7-class emotion classifier on 35,887 grayscale face images, comparing a custom CNN with MobileNetV2 transfer learning (frozen and fine-tuned). Handled class imbalance with class weighting and augmentation, selected the final model by Macro F1 (0.482) instead of accuracy, and used Grad-CAM to explain predictions. Found that fine-tuning overfit on this dataset size.
- **DeepCSAT: E-Commerce Customer Satisfaction Prediction (Deep Learning)**
  Built an ANN (Keras) to predict 1–5 CSAT scores from 85,907 support interactions. Confirmed key drivers with hypothesis tests, processed customer remarks with an NLP pipeline (TF-IDF + SVD), and handled the 69% class imbalance with class weighting. Compared 3 architectures using macro F1, and deployed the winning model as a Streamlit app with prediction confidence charts.

- **Integrated Retail Analytics & Demand Forecasting (Capstone)**
  Analyzed 421K+ weekly sales records across 45 stores, with anomaly detection (IQR, Isolation Forest), seasonal decomposition, KMeans store segmentation, and Apriori department pairing. Validated findings with hypothesis tests (t-test, ANOVA) and built a one-week-ahead demand forecast using lag features and a time-based split. The final HistGradientBoosting model reached 98.35% R^2, ahead of naive baselines (97.5%).
- **Term Deposit Prediction with Hadoop, Hive & Spark (Financial Forecasting Frontier)**
  Built a customer subscription predictor on 4,521 bank records, using Dockerized Hadoop (HDFS) and Hive for storage and HiveQL analysis, and Spark for modeling. Compared Logistic Regression, Decision Tree, and Random Forest with class-imbalance handling (tuned Random Forest: 87% accuracy). Used Spark partitioning and parallel tasks (independent jobs ran 42% faster), and built a Spark streaming pipeline with live predictions, 10-second and 1-minute windows, and watermarking for late data.
- **End-to-End MLOps Pipeline for Flight Price Prediction (Capstone)**
  Deployed a Random Forest price model as a Flask REST API with input validation, containerized it with Docker, and ran it on Kubernetes with 3 self-healing replicas. Automated weekly retraining with Airflow (with a 0.90 R^2 quality gate before deployment), built a Jenkins CI/CD pipeline that builds, tests, and deploys the API, and tracked all runs in MLflow with a Model Registry.
- **Google Play Store EDA & Dashboard**
  Conducted comprehensive exploratory data analysis on 10K+ apps and built an interactive Tableau dashboard visualizing app performance metrics, user sentiment, and market trends for strategic insights.
- **Data Summarization Pipeline (Industrial Project)**
  Leveraged Python, Selenium, and Salesforce APIs to build a complex analysis pipeline processing 500+ customer reviews daily, extracting actionable insights with 92% data accuracy.

TECHNICAL SKILLS

Languages	Python (Pandas, NumPy, Scikit-learn,Selenium,etc.), MySQL
Data Analysis & Visualization	Tableau, Power BI, Complete Advanced Excel
Machine Learning & AI	Deep Learning, NLP, LLM Fine-tuning (Hugging Face), Statistical Analysis, Web Scraping (BeautifulSoup, Selenium)
MLOps & Big Data	Docker, Kubernetes, Airflow, Jenkins, MLflow, Flask, Streamlit, Hadoop, Hive, Spark
Tools & Platforms	Git, GitHub, Anaconda Jupyter, VS Code, Impala, GPT/Gemini/Copilot/Claude for rapid prototyping
Core Competencies	ML Model Creation, Pipeline Development, Insights Generation, Process Automation, Information Management, Cross-functional Collaboration

EDUCATION

Masters in Computer Science (Machine Learning & AI Engineering) 2024 – 2026
Almabetter (Woolf University) *Mumbai, India*

Certification in Data Science and Artificial Intelligence 2023 – 2024
Develearn Institute *Mumbai, India*

Bachelor of Science in Bioanalytical Sciences 2020 – 2023
Guru Nanak Khalsa College *Mumbai, India*

AWARDS & ACHIEVEMENTS

* Market Research Society India (Best Business Impact through analytics)